EXPERIMENT 34: EVALUATION OF ANTI-ULCER AGENTS

40 STANDARD IN-COURSE MCQs

1. The primary purpose of Experiment 34 is to evaluate:

A. The diuretic potency of loop diuretics B. The anti-ulcer and gastroprotective efficacy of pharmacological agents against chemical ulcerogens C. The motor coordination effects of muscle relaxants D. The antibacterial activity of aminoglycosides

Answer: B

2. What duration of pre-experimental fasting is designated in the general protocol on Page 161?

A. 2 hours B. ~16 hours C. 48 hours D. No fasting required

Answer: B

3. Why must experimental animals be fasted for ~16 hours prior to ulcer induction?

A. To induce severe hypoglycemia B. To empty the stomach of food so the gastric mucosa is uniformly exposed to the ulcerogen C. To prevent animals from drinking water D. To destroy the gastric parietal cells

Answer: B

4. When are the anti-ulcer pre-treatment drugs administered relative to the ulcerogen?

A. 30 minutes to 1 hour before the ulcerogen B. At the exact same second in the same syringe C. 24 hours after the ulcerogen D. Immediately after sacrificing the animal

Answer: A

5. Which necrotizing chemical ulcerogen is used in Group I?

A. 0.1 N Sodium hydroxide B. Absolute ethanol (1 mL) C. Dilute acetic acid D. 10% Formalin

Answer: B

6. Which chemical ulcerogen is used in Group II?

A. 0.6 N Hydrochloric acid (HCl, 1 mL) B. 5% Acetic acid C. Sulfuric acid D. Lactic acid

Answer: A

7. What volume of ulcerogen is administered orally to each rat?

A. 0.1 mL B. 1.0 mL C. 5.0 mL D. 10.0 mL

Answer: B

8. Along which anatomical line is the harvested stomach opened for examination?

A. Lesser curvature B. Greater curvature C. Transverse duodenum D. Gastroesophageal sphincter only

Answer: B

9. Why is the excised stomach opened specifically along the greater curvature?

A. To avoid cutting through the pyloric sphincter B. The lesser curvature is the primary site of glandular mucosal folds where ulcers predominantly form C. It is impossible to cut the lesser curvature D. To prevent loss of pancreatic enzymes

Answer: B

10. What parameter is quantified from the harvested stomach contents before examination?

A. Total blood volume B. Volume, pH, and acidity of gastric secretions C. Serum bilirubin D. Bile acid concentration

Answer: B

11. What is the standard metric used to quantify the severity and extent of mucosal damage?

A. Liver weight B. Ulcer Index (UI) C. Spleen diameter D. Bleeding time

Answer: B

12. Which H_2 receptor antagonist is tested at a dose of 50 mg/kg?

A. Ranitidine B. Cimetidine C. Famotidine D. Nizatidine

Answer: B

13. What is the molecular mechanism of cimetidine in protecting against gastric ulcers?

A. Irreversible inhibition of H^+/K^+ ATPase B. Competitive antagonism of histamine H_2 receptors on gastric parietal cells, reducing acid output C. Direct chemical neutralization of luminal hydrochloric acid D. Stimulation of gastrin release

Answer: B

14. Which proton pump inhibitor (PPI) is tested in the protocol?

A. Lansoprazole B. Omeprazole (20 mg/kg or 200 mcg/mL) C. Pantoprazole D. Rabeprazole

Answer: B

15. What is the molecular mechanism of omeprazole?

A. Competitive H_1 receptor blockade B. Irreversible covalent inhibition of the H^+/K^+ ATPase enzyme (proton pump) in parietal cells C. Direct activation of EP_3 prostaglandin receptors D. Neutralization of pepsin

Answer: B

16. Which synthetic prostaglandin E_1 (PGE_1) analogue is tested at 50 mcg/kg?

A. Alprostadil B. Misoprostol C. Dinoprostone D. Carboprost

Answer: B

17. What is the dual mechanism of action of misoprostol in peptic ulcer disease?

A. Inhibits acid secretion (via EP_3 receptors) and enhances mucosal cytoprotection (mucus and bicarbonate secretion) B. Increases acid output and increases gut motility C. Destroys Helicobacter pylori and blocks gastrin D. Induces gastric vasoconstriction

Answer: A

18. Which classical nonsystemic antacid suspension is tested at 1 mL/rat?

A. Sodium bicarbonate B. Magnesium trisilicate C. Calcium carbonate D. Potassium citrate

Answer: B

19. How does magnesium trisilicate protect the gastric mucosa?

A. Irreversibly poisons parietal cells B. Chemically neutralizes gastric acid, adsorbs pepsin, and forms a protective hydrated silicon dioxide gel coating C. Activates histamine H_2 receptors D. Stimulates systemic gastrin secretion

Answer: B

20. What vehicle is administered to control animals before ulcer induction?

A. Normal saline B. Distilled water (1 mL) C. Olive oil D. Liquid paraffin

Answer: B

21. What macroscopic mucosal lesions are observed following absolute ethanol challenge in unprotected control rats?

A. Completely normal, glistening pink mucosa B. Elongated, dark red-to-black hemorrhagic bands, petechiae, and extensive mucosal necrosis C. White fibrous scar tissue D. Perforated duodenal diverticula

Answer: B

22. How does absolute ethanol induce severe acute gastric mucosal damage?

A. By stimulating massive histamine release from parietal cells B. Direct necrotizing disruption of the gastric mucosal barrier, microvascular stasis, and ischemia C. Inhibiting pepsinogen secretion D. Decreasing stomach emptying rate

Answer: B

23. How does 0.6 N HCl produce mucosal necrosis?

A. Direct acid-induced chemical coagulation necrosis of surface epithelial cells and capillary endothelial cells B. Conversion of pepsin to trypsin C. Activation of gastric amylase D. Vasodilation with mucosal edema only

Answer: A

24. What formula is used to calculate the percentage protection of an anti-ulcer drug?

A. (UI_treated / UI_control) * 100 B. [(UI_control - UI_treated) / UI_control] * 100 C. (UI_control + UI_treated) / 100 D. UI_treated * UI_control

Answer: B

25. If control rats have a mean Ulcer Index of 25.0 and omeprazole-treated rats have an Ulcer Index of 2.5, the % protection is:

A. 10% B. 50% C. 90% D. 95%

Answer: C

26. In Question 3 on Page 161, what is the scientific definition of "cytoprotection"?

A. Complete suppression of all hydrochloric acid secretion B. The property of an agent to protect gastric mucosal cells against necrotizing damage without necessarily inhibiting acid secretion C. Direct destruction of bacterial cell walls D. Surgical resection of the gastric antrum

Answer: B

27. Who originally introduced the concept of gastric cytoprotection by prostaglandins in 1979?

A. James Black B. André Robert C. Hans Selye D. Alexander Fleming

Answer: B

28. Which of the following is an endogenous mediator of gastric mucosal cytoprotection?

A. Endothelin-1 B. Prostaglandin E_2 (PGE_2) and Prostacyclin (PGI_2) C. Thromboxane A_2 D. Leukotriene C_4

Answer: B

29. How do prostaglandins enhance mucosal defense?

A. Stimulating mucus and bicarbonate secretion, increasing mucosal blood flow, and promoting epithelial restitution B. Decreasing mucosal blood flow to prevent hemorrhage C. Stimulating parietal cell acid secretion D. Increasing gastric motility to expel acid

Answer: A

30. Why do nonsteroidal anti-inflammatory drugs (NSAIDs) like aspirin and indomethacin cause gastric ulcers?

A. Direct stimulation of H^+/K^+ ATPase B. Inhibition of COX-1, depleting gastroprotective prostaglandins (PGE_2 and PGI_2) C. Blockade of histamine H_2 receptors D. Neutralization of gastric mucus

Answer: B

31. Which of the tested drugs demonstrates the most potent cytoprotective activity independent of acid suppression?

A. Cimetidine B. Misoprostol C. Sodium chloride D. Calcium carbonate

Answer: B

32. Omeprazole is a prodrug that requires which condition to convert into its active tetracyclic sulfenamide form?

A. Highly alkaline intestinal pH (> 8.0) B. An acidic environment (pH < 4.0) within parietal cell secretory canaliculi C. Hepatic glucuronidation D. Plasma esterase cleavage

Answer: B

33. Why are omeprazole capsules formulated with an enteric coating for clinical use?

A. To prevent ulceration of the tongue B. To protect the acid-labile omeprazole molecule from premature degradation by gastric acid before intestinal absorption C. To accelerate absorption in the stomach D. To enhance gastric emptying

Answer: B

34. A major side effect of cimetidine not seen with newer H_2 blockers is:

A. Aplastic anemia B. Anti-androgenic effects (gynecomastia, impotence) and potent inhibition of hepatic CYP450 enzymes C. Severe acute hypertension D. Permanent hearing loss

Answer: B

35. A major clinical contraindication to the use of misoprostol in female patients is:

A. Peptic ulcer disease B. Pregnancy (risk of uterine contractions, bleeding, and abortion) C. Asthma D. Seasonal allergic rhinitis

Answer: B

36. Which bacterium is the major causative organism for chronic human peptic and duodenal ulcer disease?

A. Escherichia coli B. Helicobacter pylori C. Streptococcus pneumoniae D. Clostridium difficile

Answer: B

37. What standard triple therapy is commonly used clinically to eradicate Helicobacter pylori?

A. Omeprazole + Clarithromycin + Amoxicillin (or Metronidazole) B. Cimetidine + Aspirin + Paracetamol C. Misoprostol + Indomethacin + Ibuprofen D. Magnesium trisilicate + Atropine + Morphine

Answer: A

38. In this experiment, what is the role of the dissecting microscope or hand lens?

A. To weigh the stomach B. To accurately identify and count microscopic petechial hemorrhages and mucosal erosion widths C. To measure the pH of gastric juice D. To titrate hydrochloric acid

Answer: B

39. If cimetidine reduces gastric ulceration produced by 0.6 N HCl more effectively than absolute ethanol, why is this?

A. HCl damage relies heavily on acid-dependent mechanisms, whereas ethanol induces necrotizing injury independent of luminal acid B. Ethanol destroys H_2 receptors permanently C. Cimetidine neutralizes ethanol chemically D. Rats do not absorb cimetidine in the presence of alcohol

Answer: A

40. The overall conclusion of Experiment 34 is that:

A. All anti-ulcer drugs share an identical mechanism of action B. Anti-ulcer drugs protect the gastric mucosa either through antisecretory mechanisms (cimetidine, omeprazole), chemical neutralization (antacids), or direct cytoprotective enhancement of mucosal defense (misoprostol) C. Only proton pump inhibitors can prevent gastric ulceration D. Gastric ulcers cannot be evaluated in rodent models

Answer: B